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PAEDIATRICS

Pre-Operative Sedation for Children

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Key Points

- Medications & dosing not stated in the summary table must be administered by an Anaesthetist.

Introduction

The aim of sedative premedication is to achieve a relaxed state with a reduction in anxiety and an increase of compliance by children and young people before medical procedures. After receiving effective premedication, patients may exhibit an altered level of consciousness, but should retain the ability to independently and continuously maintain a patent airway, follow verbal commands, and respond appropriately to tactile stimulation.

The premedication of children with severe cardio-respiratory or neurological disease is beyond the scope of this document and will require a personalised approach determined by a consultant anaesthetist.

First Line: Midazolam (oral/buccal)

For the anxious, but cooperative child, oral midazolam is often adequate.

Administer 0.5mg/kg oral midazolam (max 20mg).

Lower doses provide inadequate anxiolysis and higher doses potentially causing dysphoric reactions, ataxia and prolonged sedation, without improving anxiolysis.

Onset within 15 minutes, peak sedation around 30 minutes.

In obese children, the dose should be given according to their actual body weight up to the maximum limit of 20mg. This maximum dose should not exceeded, even for a child weighing more than 40kg.

Second Line: Combined Midazolam & Ketamine

A child who displays a high level of anxiety and emotion, or those below four years old, are more likely to be inadequately sedated by Midazolam alone.

Children with developmental delay and/or additional learning needs and/or neurodiversity often need larger doses of sedative agents. This may expose them to an increased risk of side effects such as reduced respiratory drive, altered patency of the airway, and reduction of protective reflexes. Midazolam with Ketamine is an effective combination for such children, reducing the individual drug risks yet enhancing anxiolytic and sedative effects.

Administer 0.5mg/kg midazolam + 3mg/kg ketamine orally.

This dose of Ketamine provides enhanced sedation without significantly increasing oral secretion or postoperative nausea and vomiting and with no evidence of emergence delirium.

Alternative Medications

If there is a history of paradoxical excitation with Midazolam, avoid Ketamine alone. Instead consider alternative options like Clonidine or Temazepam.

Clonidine is a Partial α2 adrenoceptor agonist with sedative, anxiolytic, analgesic properties. At the stated doses, it has minimal haemodynamic effects and no respiratory depression.

Administer 4mcg/kg oral clonidine or 2-4mcg/kg intranasal (IN) clonidine.

DRUG	ROUTE	DOSE	ONSET	NOTES	EFFECTS
Midazolam	Oral/enteral	All ages: 0.5mg/kg (max. 20mg), given 30 to 60 minutes pre-op.	30 mins	Available as oral solution or injection solution which can be given orally, but is bitter.	Sedation, anxiolysis, anterograde amnesia. Can cause a paradoxical reaction in some children.
	Buccal	3 mths-1 year: 2.5mg 1-5 years: 5mg 5-10 years: 7.5mg 10-18 years: 10mg	10-15 mins	Pre-filled syringes	
Temazepam	Oral/enteral	12 to 18 years: 10-30mg, given 1 hour pre-op.	45-60 mins	Available in liquid or tablet form	
Ketamine with Midazolam	Oral/enteral	All ages: Ketamine: 3mg/kg + Midazolam: 0.5mg/kg	20-30 mins	Injection formula can be given orally, but is bitter to taste	Enhanced sedation & analgesia. Ketamine can cause increased salivation, nystagmus & a dissociated state. Particularly useful in older autistic children.
Clonidine	Oral or Intranasal	Oral: 4cmg/kg (max. 300mcg) Intranasal: 2mcg/kg	45-60 mins		Useful analgesic properties. May cause bradycardia.

Standard Monitoring

Ensure patients maintain a patent airway, follow verbal commands and respond to tactile stimulation after premedication.

Special Circumstances: Intramuscular Ketamine

If administration by no other route is possible, but sedation is considered essential to successful induction of anaesthesia, Intramuscular (IM) ketamine is recommended.

Administer 4mg/kg produces adequate sedation within 5-10 minutes. A repeat dose of 2mg/kg may be given after 10 minutes if sedation is not adequate.

Expect Sedation can be profound, so IM ketamine must be prescribed and administered by an experienced Anaesthetist **only**, competent in Paediatric Life support. Administration should either be in the recovery room, or if on the ward, in a side room

Effects:

- Random purposeless movements or stiffness, muscle twitching, rash, and vocalisations: common and of no clinical significance.
- Tachycardia and/or hypertension: transient.
- Hypersalivation: suctioning of hypersalivation may rarely be necessary.
- Transient laryngospasm: (0.3%) Positive pressure ventilation may be required, or intubation by Rapid Sequence Induction (RSI) may be considered.
- Apnoea or respiratory depression: (0.4%) is usually transient.
- Emesis: more common in children over 8 years therefore suctioning may be necessary.
- Unpleasant emergence phenomena: more common beyond mid adolescence and will resolve in time, a quiet and low stimulation environment may assist.
- Recovery agitation: (1.4%) uncommon and transient.

Pre-Sedation

Ensure the following before administering drug:

- Obtain informed consent: The sedation plan should be explained to the caregivers & child including an explanation of the effects of ketamine (see above)
- Staff Present: Consultant Anaesthetist, ODP, staff nurse all competent in PLS.
- Senior ward staff informed of plan.
- **Resuscitation equipment must be readily available**.
- Baseline observations should include BP, HR, RR and O_2 saturation; if possible, then continuous monitoring applied, as per AAGBI.
- Facilitate/encourage non-procedural conversation prior and during administration of ketamine eg “If you were not here today where would you rather be”. This may help minimise unpleasant emergence phenomenon.
- Consider distraction techniques eg music
- Apply topical anaesthetic cream early, as it requires approximately 45 minutes to work
- Fasting (see local fasting guidelines)
- Pre-oxygenation is not recommended as may mask hypoventilation

Contraindications:

- Infants <12 months.
- Current significant respiratory illness, eg asthma, respiratory tract infection.
- Known difficult airway, history of previous airway surgery or congenital anomaly.
- Intraoral Procedures or potential intraoral bleeding such as tongue lacerations and dental procedures.
- Procedures that will stimulate the posterior pharynx.
- Cardiovascular disease where increased HR and workload are contraindicated, eg ischaemic heart disease, cardiac failure, hypertension, Wolff-Parkinson-White syndrome.
- Glaucoma or acute globe injury.
- Porphyria.
- Thyroid disease.
- Bowel obstruction.
- Psychosis.

As soon as the patient is drowsy they should be transferred to theatre.

Post-Procedure Recovery

- The child should not be discharged home until they have returned to their premorbid neurological baseline.
- Nurse in a quiet area with minimal noise and physical contact, allow dim lighting if possible, and do not stimulate prematurely.

Individualised Care Plans

There will be children with complex medical & behavioural needs who require an individualised approach. The plan should be discussed with the family and the ward informed before the day of admission.

It may be deemed appropriate by the Anaesthetist in liaison with the family, to exceed stated doses of medications in this guideline. If this is the case, then the Anaesthetist is responsible for administering the drug(s), directly monitoring and escorting the patient to Theatres. This will require additional time to be allocated to the case.

References

- Fradkin DM, Scott-Warren VL, Vashisht R, Rolfe SE (2015) An Evidence-Based Guideline for the Pre-Operative Sedation of Children. J Pediatr Neonatal Care 2(6): 00095.
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